



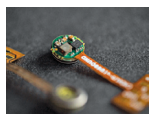
NEW PRODUCTS

MAKE IN INDIA

COMPONENTS

Compact audio module

USound has released an ultra-compact audio module that can enhance the performance of TWS and OTC hearing aids. It uses physical frequency crossover technology with a sound chamber



and pipe structures, making it simpler to integrate into a product. The audio

module is available in two versions: a two-way audio system and a full-range ultra-compact audio module. Various electronic elements can be directly embedded into the substrate. It also has extended bandwidth, allowing HD and ultrasound applications. Manufacturers can easily integrate the module in earphones and other hearables by changing the venting volume to achieve a different sound experience.

USound

<https://www.usound.com>

High-side switch controller

STMicroelectronics has revealed VNF1048F, an automotive high-side switch controller for a power MOS-



FET. It integrates advanced system protection and diagnostic features

with ST's I2-t silicon-fuse technology. The VNF1048F is compatible with 12V, 24V, and 48V automotive power subsystems (board nets) due to integration with I2-t protection. It consists of a serial peripheral interface (SPI) to enable the host microcontroller to support monitor system protection and diagnostic features. It also provides thermal protection for external high-side MOSFET, desaturation shutdown, and I2-t fuse settings, including under-

voltage and over-temperature protection. The switch has built-in high-precision digital current sensing.

ST Microelectronics

<https://www.st.com>

SoC for 4K smart TVs

Pentonic 700 by MediaTek is a smart TV system-on-chip (SoC) with a powerful AI processing engine. The



SoC can be used to design new generation of smart 4K TVs with 120Hz display.

The SoC has an AI processing unit (APU) which offers advanced features and superior image quality. It supports variable refresh rate (VRR) of up to 144Hz for gaming applications and an HDR10+ Adaptive technology, which optimises the viewing experience based on room lighting. The SoC can also enable voice assistant features and smart home device management. The Pentonic 700 supports the latest VVC H.266 standard, including HEVC, AV1, and AVS3 standards, to meet the broadcasters and streaming providers' latest requirements. The smart TV SoC is suitable for use in applications such as gaming industry, sports broadcasting, and high-end consumer TVs.

MediaTek

www.mediatek.com

Ultra-low-light camera

RouteCAM CU20 for e-con systems is an HDR GigE ultra-low-light camera with network synchronisation capabil-



ity. The camera is based on Sony's STARVIS IMX462 image sensor, which is known for its excellent low-light imaging. The

RouteCAM CU20 is a power over ethernet (PoE) camera, which can transfer both power and data at distances up to 100m over a single GigE interface. The camera can easily be integrated with all image processing platforms, including X86 processors and ARM processors, making it easy to add to your existing infrastructure without extra resources.

e-con systems

<https://www.e-consystems.com>

Battery charge management IC

Halo Microelectronics' HL7090 is a highly integrated 900mA Li-ion battery charging management IC. It is suitable



for single-cell Li-ion and single-cell Li-polymer batteries used in vari-

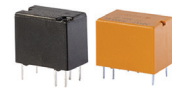
ous IoT and wearable applications. The battery charging management IC contains one switching charger, two linear chargers, and a full-range programmable charge parameters through an I2C compatible interface. The charge management IC features trickle charge, constant current charge (CC), and constant voltage charge (CV) for safely charging the battery. It comes in a small package measuring just 3mm x 3mm, which makes it suitable for use in wearables and hearables, including wireless earbuds.

Halo Microelectronics

<https://halomicro.com>

Relays for low-current switching

CUI Devices has launched a new line of signal relays for low-level current switching applications. The relays offer fast switching speed with a



maximum switching current of up to 2A in SPDT and 3A in

DPDT contact formats. They come in a compact package and have a carrying coil power ratings from 150mW